

Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
7.	(a)			a species that accepts H^+ / releases OH^- in a reversible reaction	1			1		
	(b)			a buffer maintains a constant pH upon addition of small amounts of acid or base (1) any two of following for (1) each $\text{NH}_4^+ \rightleftharpoons \text{NH}_3 + \text{H}^+$ - addition of H^+ shifts equilibrium to left / addition of OH^- reacts with H^+ which shifts equilibrium to right to replace the H^+ - salt dissociates completely releasing ammonium ions and ammonia sets up a reversible reaction	3			3		1
	(c)	(i)		pH at half volume for neutralisation = 9.3 (allow 9.25 to 9.32) (1) $K_a = 10^{-9.3} = 5.0 \times 10^{-10}$ (allow $4.79\text{-}5.62 \times 10^{-10}$) (1) allow ecf only for misread pH in the range 9.15 to 9.45		1		2	1	1
		(ii)		4-nitrophenol as pH range lies on the steep / vertical part of the curve		1		1		1
				Question 7 total	4	2	1	7	1	3